

Lab 8: Cellular Genetics — Mitosis & Meiosis

BIOL-1

Name: _____ Date: _____

Overview

Cells must divide for an organism to grow, to repair damaged tissues, and to reproduce. Eukaryotic cells have two fundamental methods of division: **Mitosis** (for creating identical diploid somatic cells) and **Meiosis** (for creating genetically unique haploid gametes). In this comprehensive lab, you will compare these two processes. You will calculate the relative time cells spend in each mitotic phase by observing real tissue, diagram the critical stages of division, and model how meiosis generates the raw material for evolution: genetic diversity.

Learning Objectives

By the end of this lab, you will be able to:

1. Identify the phases of the cell cycle (Interphase, Prophase, Metaphase, Anaphase, Telophase/Cytokinesis) in actively dividing plant and animal cells.
2. Calculate the approximate duration of each mitotic phase based on cell counts from a tissue sample.
3. Model the physical movement of chromosomes during Meiosis I and Meiosis II.
4. Demonstrate how *crossing over* and *independent assortment* create unique genetic combinations.
5. Contrast the biological purposes and outcomes of mitosis (2 diploid clones) versus meiosis (4 haploid unique cells).

Materials

- Review Packet (this document)
 - Compound Microscope & prepared slides (e.g., *Allium* (Onion) root tip, Whitefish blastula) — or *high-resolution reference images provided by instructor*
 - Chromosome Modeling Kit (Pop beads, pipe cleaners, or colored clay)
 - Colored pencils
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Part 1: The Mitotic Cell Cycle

The cell cycle consists of Interphase (G₁, S, G₂) and the M-phase (Mitosis and Cytokinesis). Mitosis is a continuous process, but we divide it into distinct phases for study based on the movement of the chromosomes.

1A. Visualizing the Phases

Act as a scientific illustrator. Below, draw detailed representations of an animal cell going through the cell cycle. Assume the cell has a diploid number of **2n = 4**.

In your drawings, carefully represent and label: **Chromatin, Chromosomes (Sister Chromatids), the Nuclear Envelope, Spindle Fibers, Centrioles, and the Cleavage Furrow.**

Box 1: Interphase (G₂)

Show chromatin inside an intact nucleus.

Box 2: Metaphase

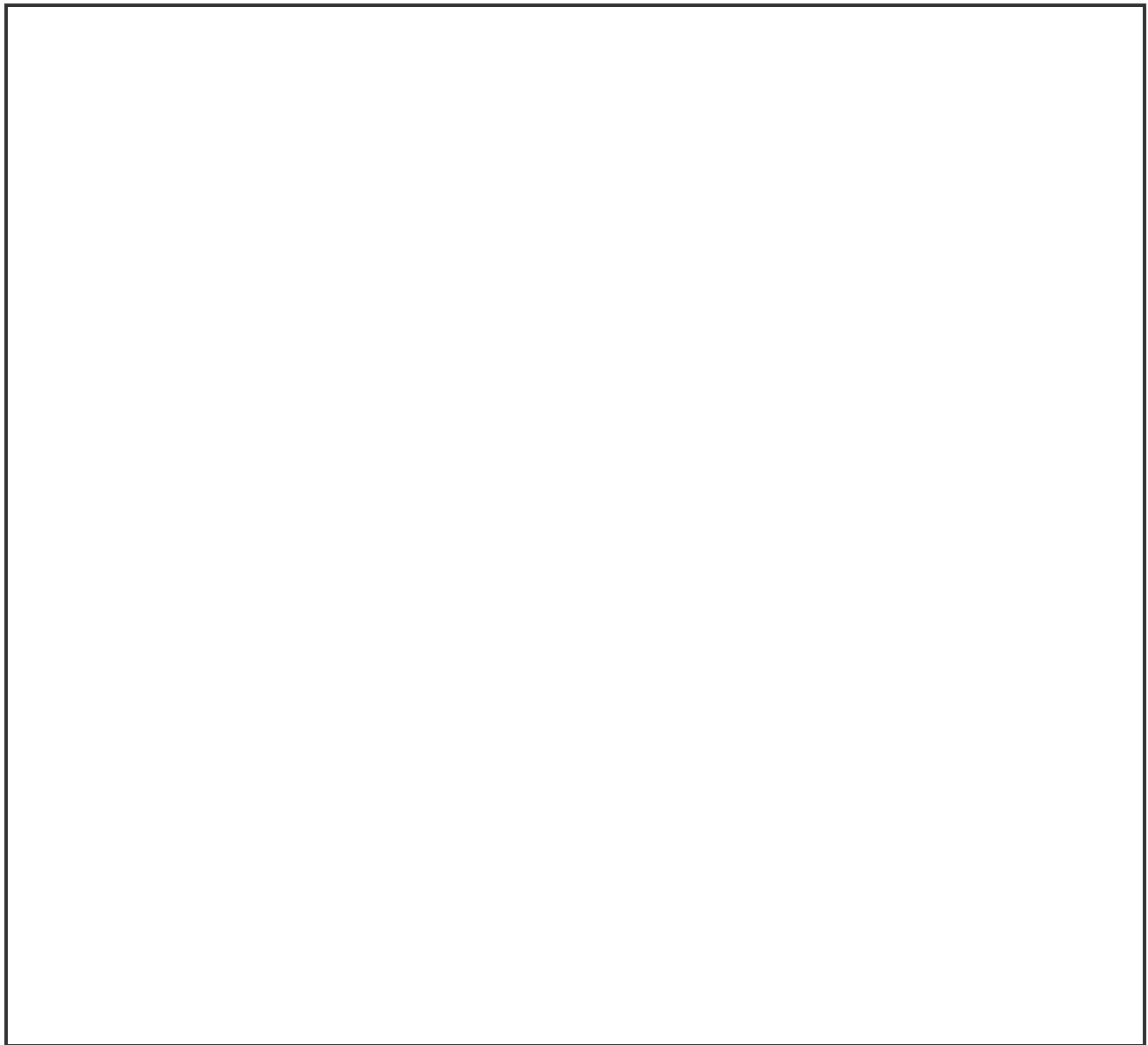
Show chromosomes lined up at the equator.

Box 3: Anaphase

Show sister chromatids being pulled apart.

Box 4: Telophase & Cytokinesis

Show two forming nuclei and the cell pinching in half.



1B. Calculating Time Spent in Phases

The root tip of an onion is a region of rapid growth (a meristem). By counting the number of cells in each phase in a snapshot of this tissue, we can estimate how long a cell actually spends in that phase.

- *Assumption:* If 10% of the observed cells are in Prophase, then Prophase takes approximately 10% of the total cell cycle time.
- *Given:* The onion root tip cell cycle takes exactly **24 hours (1,440 minutes)** to complete.

Procedure: Look at the provided slide (or micrograph image). Count 100 random cells and categorize them by their current phase. Add your data to the table. (Sample data is provided if you cannot observe a slide).

Sample Data: Interphase (72 cells), Prophase (14 cells), Metaphase (6 cells), Anaphase (4 cells), Telophase (4 cells).

1. Complete the Data Table:

Phase	Number of Cells Counted	Percentage of Total Cells	Time in Phase (Minutes)
Interphase			
Prophase			
Metaphase			
Anaphase			
Telophase			
TOTAL	100	100%	1,440 min

2. Based on your calculations, which phase takes the longest? Why does this make sense biologically?

3. Which mitotic phase is the shortest? What physical event is happening during this fleeting moment?

Part 2: Meiosis and Genetic Diversity

While mitosis produces identical clones, meiosis is a reduction division that cuts the chromosome number in half (Diploid $2n \rightarrow$ Haploid n), returning to the diploid state only upon fertilization. Meiosis introduces immense genetic variety through two specific mechanisms in Meiosis I.

Procedure: Use your chromosome kits to model a cell where $2n = 4$. Use two long chromosome models (one Red for maternal, one Blue for paternal) and two short chromosome models (one Red, one Blue).

2A. Crossing Over (Prophase I)

1. Put your cell into Interphase and replicate the DNA (create X-shaped sister chromatids).
2. Enter Prophase I. Bring the homologous pairs together to form **tetrads** (Long Red with Long Blue; Short Red with Short Blue).
3. **Simulate Crossing Over:** Physically break off a small bottom piece of the Long Red chromatid and swap it with the exact corresponding bottom piece of the Long Blue chromatid.

4. Describe the visual result of your crossing over simulation. Are the sister chromatids still identical to each other?

2B. Independent Assortment (Metaphase I)

1. Move the tetrads to the cell equator for Metaphase I.
2. How did you line them up?
 - *Option A:* Both Red chromosomes on the left, both Blue on the right.
 - *Option B:* Long Red on the left, but Short Red on the right.
3. **Simulate Anaphase I and Telophase I** by separating the homologous pairs into two new cells based on how you lined them up.

5. By the end of Meiosis I, have the sister chromatids separated yet?

6. If a human cell has 23 homologous pairs, how many different ways can they randomly align at the equator during Metaphase I? (Hint: The formula is 2^n , where n is the haploid number). Calculate this number.

2C. Meiosis II

1. Take the two cells from the end of Meiosis I. Line the chromosomes up single file (Metaphase II).
2. Separate the sister chromatids (Anaphase II) to create four distinct cells (Telophase II).

Draw the genetic contents of the four final haploid cells you just created with your models. Show the specific combinations of Red and Blue (and the crossed-over segments) in each cell.

Part 3: Mitosis vs. Meiosis Comparison

7. Complete the comparison table.

Characteristic	Mitosis	Meiosis
Number of Divisions		
Number of Daughter Cells Created		
Are daughter cells identical to parent?		
Ploidy of daughter cells (Diploid or Haploid?)		
Biological Purpose (Why do this?)		

8. If a dog's somatic (body) muscle cell contains 78 chromosomes, how many chromosomes will be found in a dog's sperm cell? Explain why this reduction is strictly necessary for sexual reproduction.